Rebecca Wilson

Applied Research Engineer | Amsterdam, NL | Open to relocation
Email: rebeccawilson0401@gmail.com | LinkedIn: Rebecca Wilson | Engineering Portfolio

Summary

Applied Research Engineer with hands-on experience designing, building and running test programmes for laser rail-cleaning systems, from bench-scale test rigs to live-track field trials. Technical lead for laser ablation testing at Laser Precision Solutions, building fixtures, instrumenting rigs with optical and laser-based sensors and automating data analysis in Python. Aeronautical engineering graduate (University of Glasgow). English (native) and Dutch (professional working proficiency).

Key Skills

- **Testing & Instrumentation:** Test design and protocols, fixture design, tribology and adhesion testing, sensor integration, root-cause analysis, field trials
- **CAD & Fabrication:** SolidWorks, Fusion 360, 3D printing, machining (lathe and mill), soldering and wiring
- **Software & Data:** Python (pandas, OpenCV), TypeScript, MATLAB, Raspberry Pi, data acquisition and signal processing
- **Communication:** Test procedures, client-facing reports, technical presentations, guiding a data analyst

Experience

Applied Research Engineer - Laser Precision Solutions (LPS) | Aug 2026 - Present

- Technical lead for laser ablation testing: build test fixtures from hand tools and 3D-printed parts, source components, and instrument rigs with optical and laser-based sensors to evaluate tribology and surface adhesion after laser treatment.
- Standardised a laser cleaning test method that reliably removes 25 µm of contamination in a single pass, now extending testing toward 50 µm.
- Technical lead for a 4-day live-track field trial with Network Rail (UK) in December 2026, leading a team of 2-3.
- Leading concept design of a compact compressor and chiller package for a tram deployment, targeting a 30-50% smaller footprint than the train-based setup.
- Automate results processing and out-of-spec flagging in Python, write formal test procedures and client reports, and guide a data analyst day to day.

Application Engineer - Laser Precision Solutions (LPS) | Apr 2025 - Aug 2026

- Supported LaserTrain deployments for the Massachusetts Bay Transportation Authority (MBTA), Metro-North Railroad, Long Island Rail Road and Staten Island Railway, from installation and commissioning to testing and handover.
- Key technical contact between customers and internal engineering, product and software teams, coordinating multiple projects across distributed teams.
- Supported customer visits, demonstrations and industry conferences, including InnoTrans and RailTech Europe 2026, and hosted a Young Rail Professionals visit.

Selected Projects

Automated Camera and Image Analysis System - ring test rig, LPS

- Built a Raspberry Pi camera system with 3D-printed mounts that detects 16 ArUco markers on a rotating ring and captures an image automatically when each marker is centred and still, with locked exposure and white balance for repeatable images.
- Wrote Python (OpenCV) analysis that corrects for lighting drift against a reference card and measures contamination coverage against each marker's clean baseline, combined with gauge thickness readings, to support fluence testing across the laser cleaning band.

Laser Cleaning Validation - European rail operator

- Designed a test method using a rotating test rig and rail tribometer to measure friction before and after cleaning at average and worst-case contamination thicknesses; confirmed removal at real-world speeds with friction restored to near clean rail.

Education

BEng Aeronautical Engineering | University of Glasgow | Sep 2019 - Jun 2024

- Key areas: propulsion, aircraft design, control systems, aerodynamics.